



LAB TASKS ASSIGNMENT

PROGRAMMING FUNDAMENTALS



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LAB. TASK LIST #1

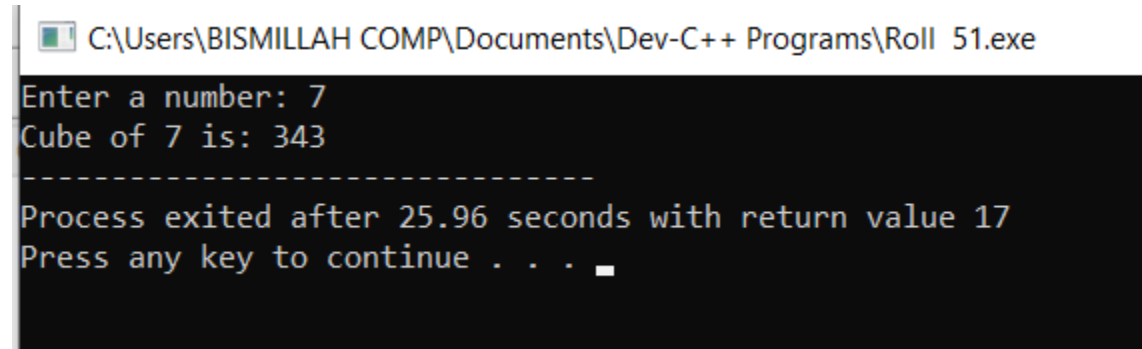
Task No. 1) Write a program that takes an integer number input and prints cube of that number.

Sample Output:

Enter a number: 5

Cube of 5 is: 125

```
#include<stdio.h>
void main()
{
    printf("Enter a number: ");
    int cube;
    scanf("%d",&cube);
    printf("Cube of %d is: %d",cube,(cube*cube*cube));
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter a number: 7
Cube of 7 is: 343
-----
Process exited after 25.96 seconds with return value 17
Press any key to continue . . .
```

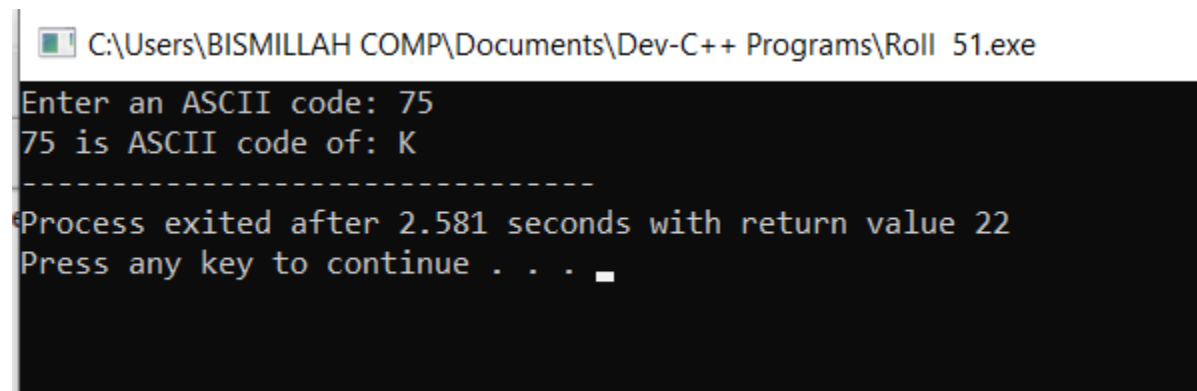
Task No. 2) Write a program that takes an ASCII code (in the range of 0 to 255) as input and prints the character which is the representation of given ASCII code.

Sample Output:

Enter an ASCII code: 67

67 is ASCII code of: C

```
#include<stdio.h>
void main()
{
    int ascii;
    printf("Enter an ASCII code: ");
    scanf("%d",&ascii);
    printf("%d is ASCII code of: %c",ascii,ascii);
}
```

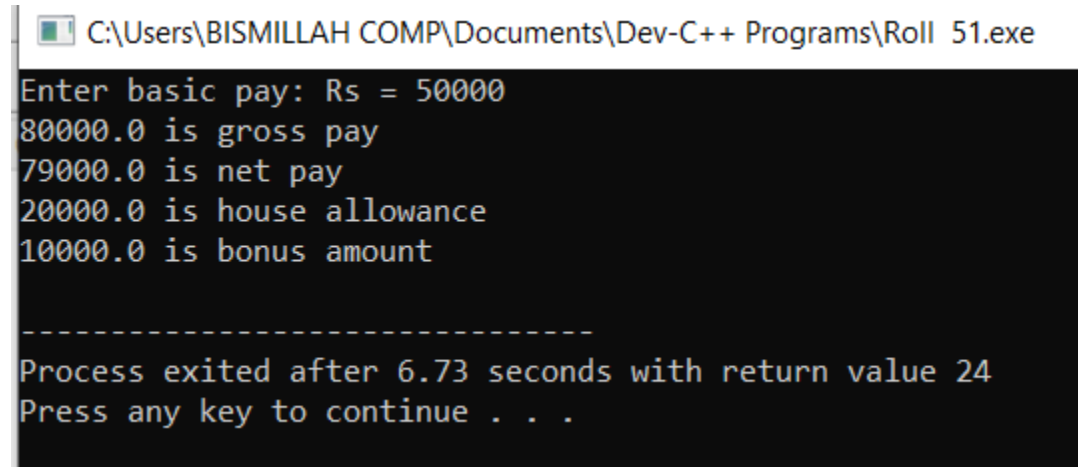


```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter an ASCII code: 75
75 is ASCII code of: K
-----
Process exited after 2.581 seconds with return value 22
Press any key to continue . . .
```

Task No. 3) Ahmed's basic salary is input through the keyboard. His conveyance allowance is 20% of basic salary, house rent allowance is 40% of basic salary and loan installment is Rs. 1000/-. Write a program to calculate his gross and net salary.

```
#include<stdio.h>
void main()
{
    float basic_salary, gross_salary, net_salary, housing_allowance, bonus_amount,
    loan_installment=1000;
    printf("Enter basic pay: Rs = ");
    scanf("%f",&basic_salary);

    housing_allowance=basic_salary*0.4;
    bonus_amount=basic_salary*0.2;
    gross_salary=basic_salary+housing_allowance+bonus_amount;
    net_salary=gross_salary-loan_installment;
    {
        printf("%.1f is gross pay\n",gross_salary);
        printf("%.1f is net pay\n",net_salary);
        printf("%.1f is house allowance\n",housing_allowance);
        printf("%.1f is bonus amount\n",bonus_amount);
    }
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter basic pay: Rs = 50000
80000.0 is gross pay
79000.0 is net pay
20000.0 is house allowance
10000.0 is bonus amount

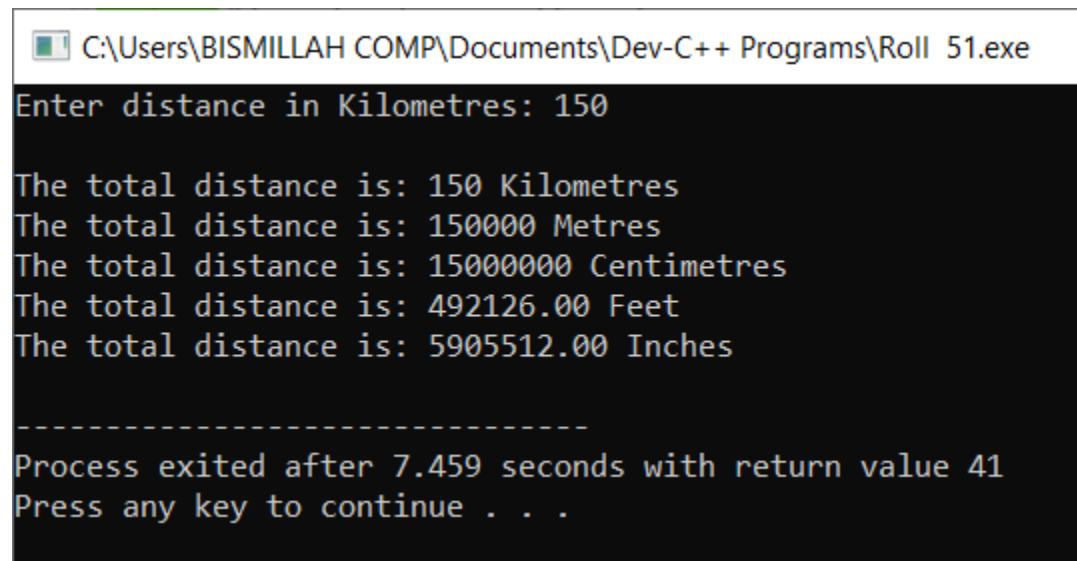
-----
Process exited after 6.73 seconds with return value 24
Press any key to continue . . .
```

Task No. 4) The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters.

```
#include<stdio.h>
void main()
{
    int kilometers, meters, centimeters;
    printf("Enter distance in Kilometres: ");
    scanf("%d",&kilometers);

    float feet,inches;

    meters=kilometers*1000;
    centimeters=meters*100;
    feet=kilometers*3280.84;
    inches=feet*12;
    {
        printf("\nThe total distance is: %d Kilometres\n",kilometers);
        printf("The total distance is: %d Metres\n",meters);
        printf("The total distance is: %d Centimetres\n",centimeters);
        printf("The total distance is: %.2f Feet\n",feet);
        printf("The total distance is: %.2f Inches\n",inches);
    }
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter distance in Kilometres: 150

The total distance is: 150 Kilometres
The total distance is: 150000 Metres
The total distance is: 15000000 Centimetres
The total distance is: 492126.00 Feet
The total distance is: 5905512.00 Inches

-----
Process exited after 7.459 seconds with return value 41
Press any key to continue . . .
```

Task No. 5) Write a program that takes width and length of a rectangle and then calculate and display the area and perimeter of the rectangle.

Sample Output:

Enter width of the rectangle: 10

Enter length of the rectangle: 5

Area of rectangle having width 10 and length 5 is:

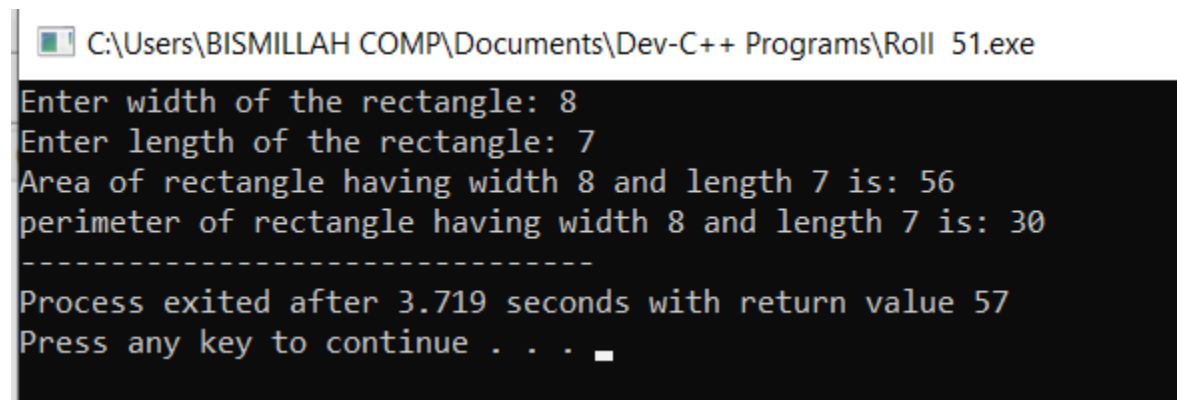
Perimeter of rectangle having width 10 and length 5 is:

Hint: Area of rectangle = length x width and Perimeter of rectangle = 2 (length + width)

```
#include<stdio.h>
void main()
{
    int width, length, area, perimeter;
    printf("Enter width of the rectangle: ");
    scanf("%d",&width);
    printf("Enter length of the rectangle: ");
    scanf("%d",&length);

    area = width*length;
    perimeter = 2*(width+length);

    printf("Area of rectangle having width %d and length %d
is: %d\n",width,length,area);
    printf("perimeter of rectangle having width %d and length %d
is: %d",width,length,perimeter);
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter width of the rectangle: 8
Enter length of the rectangle: 7
Area of rectangle having width 8 and length 7 is: 56
perimeter of rectangle having width 8 and length 7 is: 30
-----
Process exited after 3.719 seconds with return value 57
Press any key to continue . . .
```

Task No. 6) Write a program that takes temperature in Fahrenheit and converts it in Celsius degree and display it on screen.

Sample Output:

**Enter temperature in Fahrenheit : 67
67.00 Fahrenheit = 19.44 Celsius**

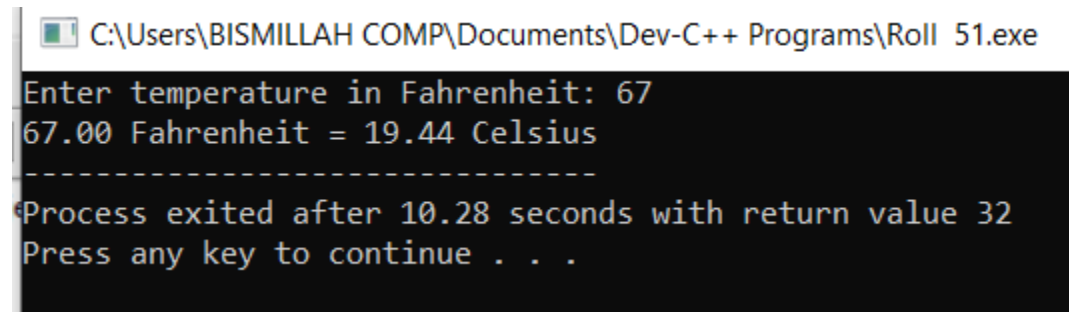
Hint: Formula $(32^{\circ}\text{F} - 32) \times 5/9 = 0^{\circ}\text{C}$

Note: $32^{\circ}\text{F} = 0^{\circ}\text{C}$

```
#include<stdio.h>
void main()
{
    float fahrenheit, celsius;
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f",&fahrenheit);

    celsius=(fahrenheit-32)*5/9;

    printf("%.2f Fahrenheit = %.2f Celsius",fahrenheit,celsius);
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter temperature in Fahrenheit: 67
67.00 Fahrenheit = 19.44 Celsius
-----
Process exited after 10.28 seconds with return value 32
Press any key to continue . . .
```


LAB. TASK LIST #2

Task No. 1) If the marks obtained by a student in five different subjects are input through the keyboard, find out the aggregate marks, percentage and grade obtained by the student. Assume that the maximum marks that can be obtained by a student in each subject is 100

```
#include <stdio.h>
void main()
{
    int sub1, sub2, sub3, sub4, sub5;
    printf("Enter obtained marks of all subjects: ");
    scanf("%d %d %d %d %d",&sub1,&sub2,&sub3,&sub4,&sub5);
    {
        if(sub1>100 && sub2>100 && sub3>100 && sub4>100 && sub5>100);
        printf("\nYou obtained %d marks out of
500\n",sub1+sub2+sub3+sub4+sub5);
        printf("Your obtained percentage is %d\n",(sub1+sub2+sub3+sub4+sub5)/5);
        if((sub1+sub2+sub3+sub4+sub5)/5>=90)
            printf("A1 Grade");
        else if((sub1+sub2+sub3+sub4+sub5)/5>=80)
            printf("A Grade");
        else if((sub1+sub2+sub3+sub4+sub5)/5>=70)
            printf("B Grade");
        else if((sub1+sub2+sub3+sub4+sub5)/5>=60)
            printf("C Grade");
        else if((sub1+sub2+sub3+sub4+sub5)/5>=50)
            printf("D Grade");
        else
            printf("You're Fail");
    }
}
```

 C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe

Enter obtained marks of all subjects: 67 89 87 65 78

You obtained 386 marks out of 500

Your obtained percentage is 77

B Grade

Process exited after 10.48 seconds with return value 7

Press any key to continue . . . ■

Task No. 2) Any character is entered through the keyboard, write a program to determine whether the character entered is a capital letter, a small case letter, a digit or a special symbol.

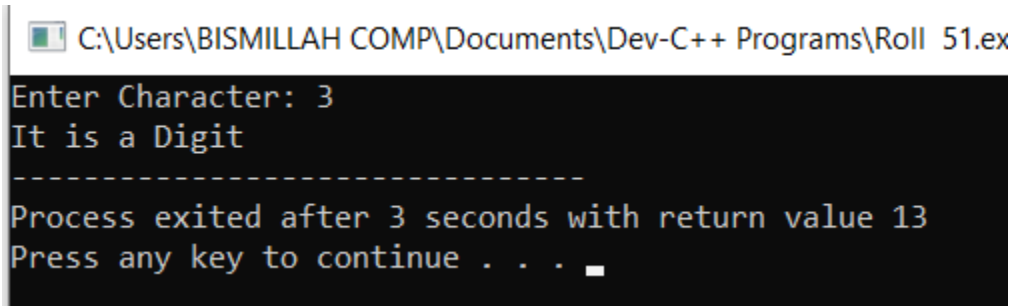
```
#include <stdio.h>
void main()
{
    char ASCII;
    printf("Enter Character: ");
    scanf("%c",&ASCII);

    if(ASCII>=65 && ASCII<=90)
        printf("It is a Capital Letter");

    else if (ASCII>=97 && ASCII<=122)
        printf("It is a Small Case Letter");

    else if (ASCII>=48 && ASCII<=57)
        printf("It is a Digit");

    else
        printf("It is a Special Symbol");
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.ex
Enter Character: 3
It is a Digit
-----
Process exited after 3 seconds with return value 13
Press any key to continue . . .
```

Task No. 3) A library charges a fine for every book returned late. For first 5 days the fine is Rs. 5, for 6-10 days fine is Rs. 10 and above 10 days fine is Rs. 25. If you return the book after 30 days your membership will be cancelled. Write a program to accept the number of days the member is late to return the book and display the fine or the appropriate message.

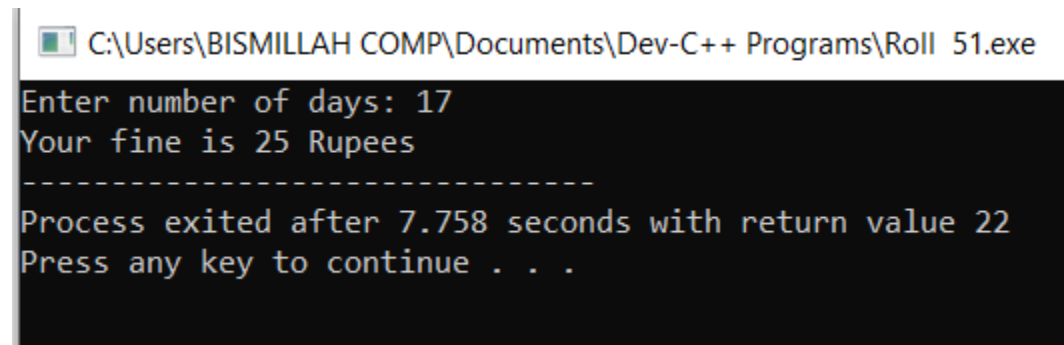
```
#include <stdio.h>
void main()
{
    int days;
    printf("Enter number of days: ");
    scanf("%d",&days);

    if(days<6)
        printf("Your fine is 5 Rupees");

    else if(days<11)
        printf("Your fine is 10 Rupees");

    else if(days<31)
        printf("Your fine is 25 Rupees");

    else if(days>30)
        printf("Your no longer a member");
}
```



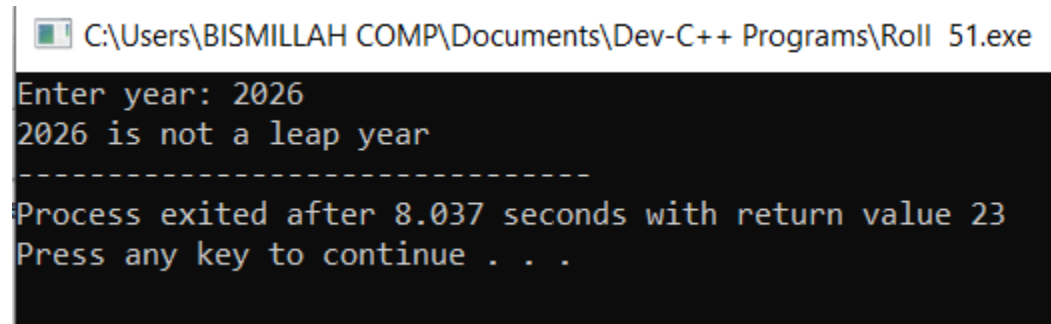
```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter number of days: 17
Your fine is 25 Rupees
-----
Process exited after 7.758 seconds with return value 22
Press any key to continue . . .
```

Task No. 4) Any year is input through the keyboard. Write a program to determine whether the year is a leap year or not. (Hint: Use the % (modulus) operator)

```
#include<stdio.h>
void main()
{
    int leap_year;
    printf("Enter year: ");
    scanf("%d",&leap_year);

    if(leap_year % 4 == 0)
        printf("%d is a leap year",leap_year);

    else
        printf("%d is not a leap year",leap_year);
}
```



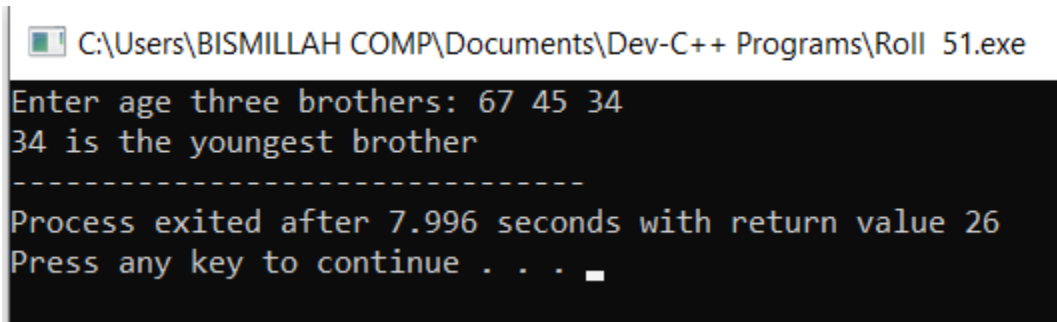
```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter year: 2026
2026 is not a leap year
-----
Process exited after 8.037 seconds with return value 23
Press any key to continue . . .
```

Task No. 5) If the ages of three brothers are input through the keyboard, write a program to determine the youngest of the three using nested if() statement without using logical operators.

```
#include<stdio.h>
void main()
{
    int bro1,bro2,bro3;
    printf("Enter age three brothers: ");
    scanf("%d %d %d",&bro1,&bro2,&bro3);

    if(bro1<bro2)
        if(bro1<bro3)
            printf("%d is the youngest brother",bro1);
        else
            printf("%d is the youngest brother",bro3);

    else if(bro2<bro3)
        printf("%d is the youngest brother",bro2);
    else
        printf("%d is the youngest brother",bro3);
}
```

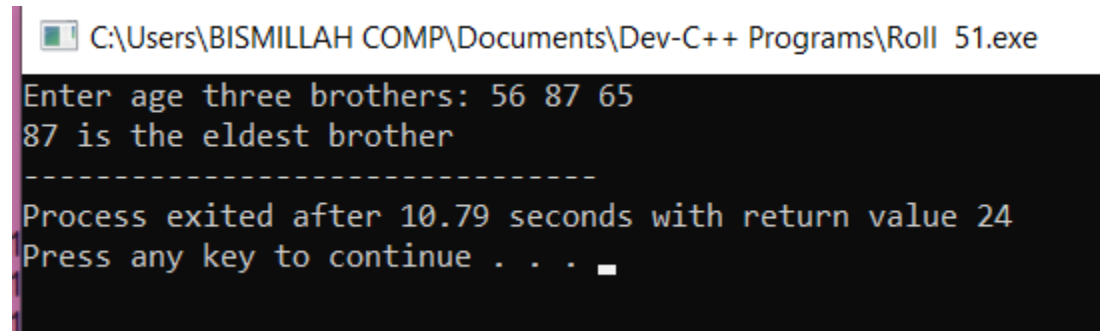


```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter age three brothers: 67 45 34
34 is the youngest brother
-----
Process exited after 7.996 seconds with return value 26
Press any key to continue . . .
```

Task No. 6) If the ages of three brothers are input through the keyboard, write a program to determine the eldest of the three using logical operators.

```
#include<stdio.h>
void main()
{
    int bro1,bro2,bro3;
    printf("Enter age three brothers: ");
    scanf("%d %d %d",&bro1,&bro2,&bro3);

    if(bro1>bro2 && bro1>bro3)
        printf("%d is the eldest brother",bro1);
    else if(bro2>bro1 && bro2>bro3)
        printf("%d is the eldest brother",bro2);
    else
        printf("%d is the eldest brother",bro3);
}
```



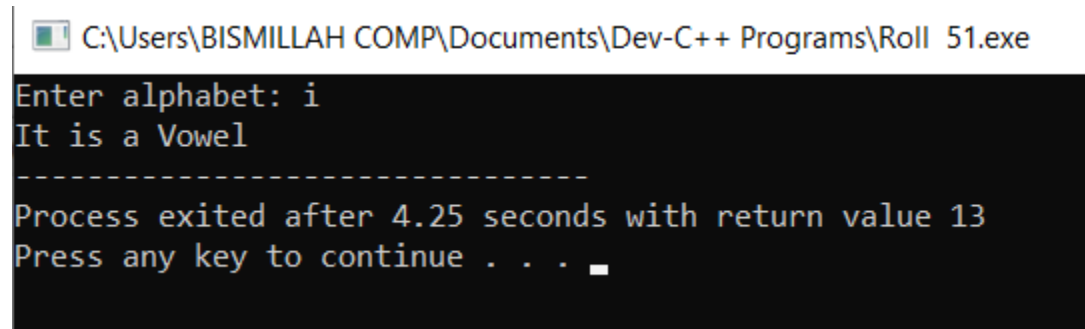
```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter age three brothers: 56 87 65
87 is the eldest brother
-----
Process exited after 10.79 seconds with return value 24
Press any key to continue . . .
```

Task No. 7) Any character is entered through the keyboard, write a program to determine whether the character entered is a vowel or consonant.

```
#include <stdio.h>
void main()
{
    char alphabet;
    printf("Enter alphabet: ");
    scanf("%c",&alphabet);

    if(alphabet == 'a' || alphabet == 'e' || alphabet == 'i' || alphabet == 'o' || alphabet
    == 'u' || alphabet == 'A' || alphabet == 'E' || alphabet == 'I' || alphabet == 'O' ||
    alphabet == 'U' )
    {
        printf("It is a Vowel");
    }

    else
    {
        printf("It is a Consonant");
    }
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter alphabet: i
It is a Vowel
-----
Process exited after 4.25 seconds with return value 13
Press any key to continue . . .
```


LAB. TASK LIST #3

Task No. 1) Write a program to generate the table of given number input through the keyboard.

Enter any table number: 5

Table of 5

=====

5 x 1 = 5

..

..

5 x 10 = 50

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int table, loop;
```

```
    printf("Enter any table number: ");
```

```
    scanf("%d",&table);
```

```
    printf("Table of %d\n",table);
```

```
    printf("=====\n");
```

```
    for(loop=1; loop<=12; loop++)
```

```
        printf("%d x %d = %d\n",table,loop,(table*loop));
```

```
}
```

C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe

Enter any table number: 7

Table of 7

=====

7 x 1 = 7

7 x 2 = 14

7 x 3 = 21

7 x 4 = 28

7 x 5 = 35

7 x 6 = 42

7 x 7 = 49

7 x 8 = 56

7 x 9 = 63

7 x 10 = 70

7 x 11 = 77

7 x 12 = 84

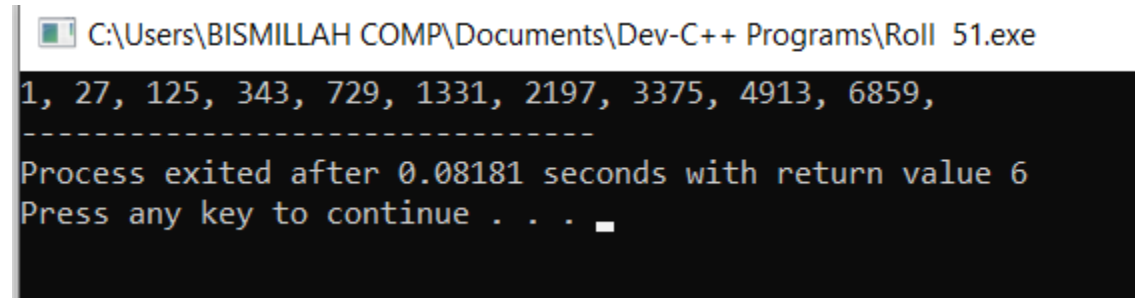
Process exited after 4.372 seconds with return value 12

Press any key to continue . . .

Task No. 2) Write a program to generate the following series. The output should look like as given below:

1, 27, 125, 343, 729, 1331, 2197, 3375, 4913, 6859,

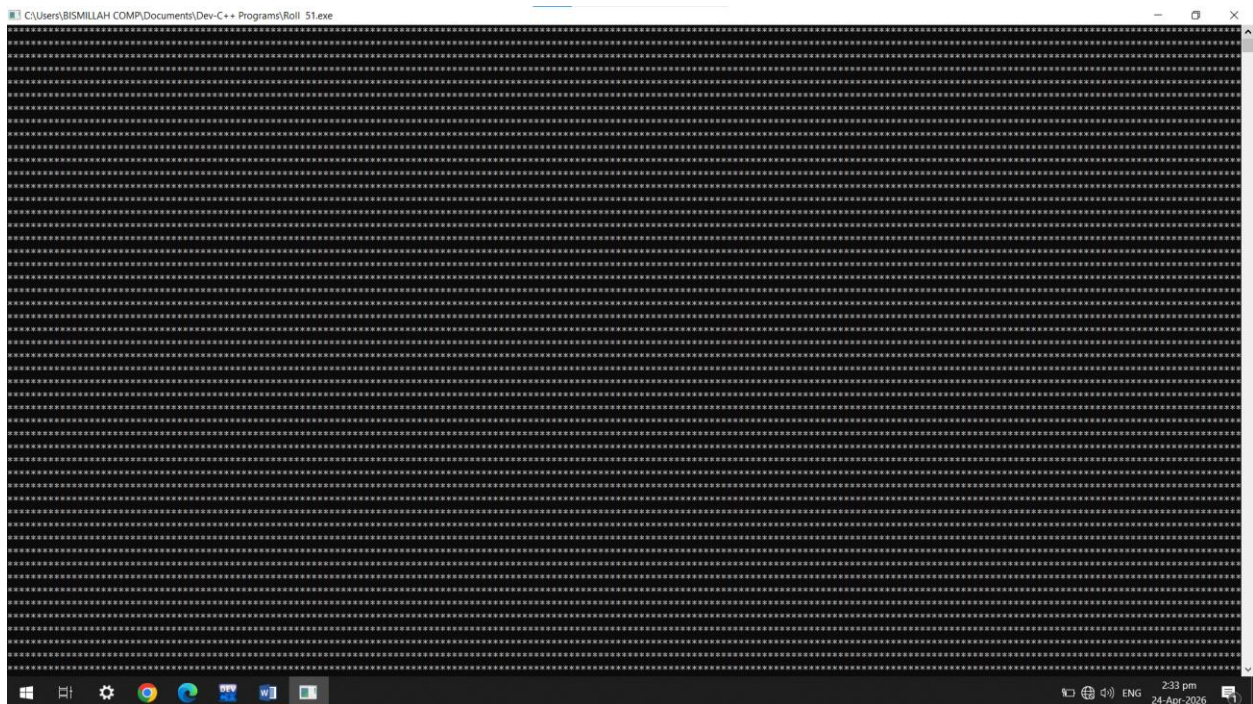
```
#include<stdio.h>
void main()
{
    int cubes;
    for(cubes=1; cubes<=19; cubes=cubes+2)
        printf("%d, ",(cubes*cubes*cubes));
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
1, 27, 125, 343, 729, 1331, 2197, 3375, 4913, 6859,
-----
Process exited after 0.08181 seconds with return value 6
Press any key to continue . . .
```

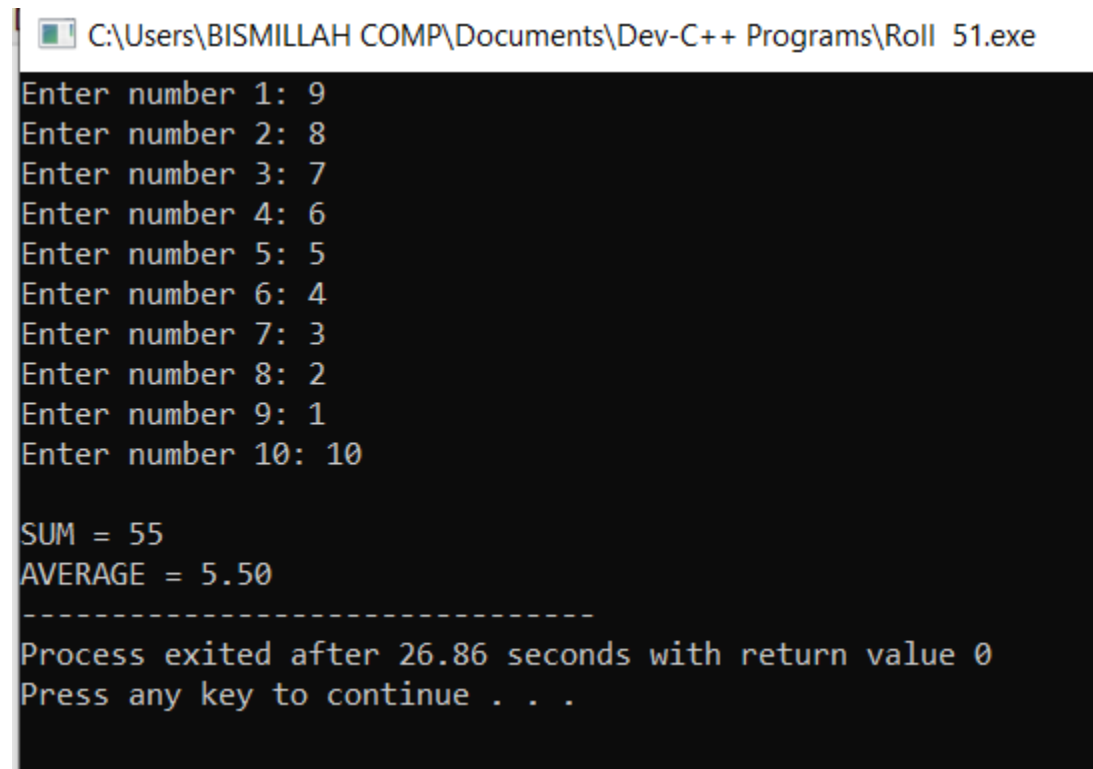
Task No. 3) Write a program to fill the entire screen with an asterisk (*) character.

```
#include<stdio.h>
void main()
{
    int star;
    for(star=1; star<=11000; star++)
        printf("*");
}
```



Task No. 5) Write a program that gets 10 numbers as input from the user and calculates and displays their SUM and AVERAGE on screen.

```
#include <stdio.h>
int main() {
    int num, i, sum = 0;
    float avg;
    for(i = 1; i <= 10; i++) {
        printf("Enter number %d: ", i);
        scanf("%d", &num);
        sum += num;
    }
    avg = sum / 10.0;
    printf("\nSUM = %d\n", sum);
    printf("AVERAGE = %.2f", avg);
    return 0;
}
```



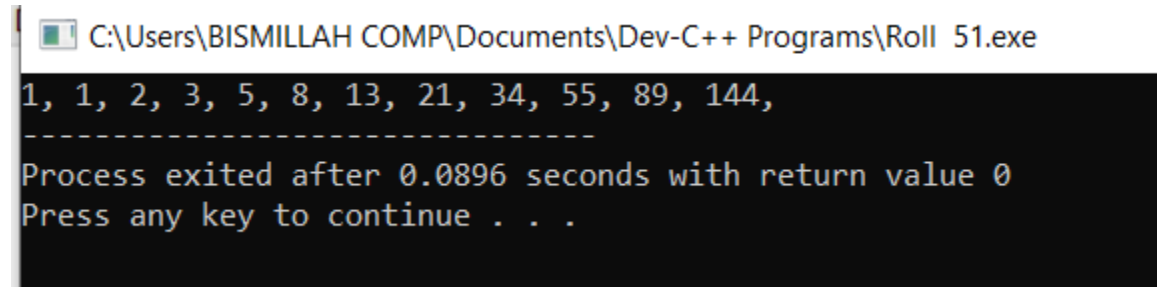
```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter number 1: 9
Enter number 2: 8
Enter number 3: 7
Enter number 4: 6
Enter number 5: 5
Enter number 6: 4
Enter number 7: 3
Enter number 8: 2
Enter number 9: 1
Enter number 10: 10

SUM = 55
AVERAGE = 5.50
-----
Process exited after 26.86 seconds with return value 0
Press any key to continue . . .
```

Task No. 6) Write a program to generate the Fibonacci series. The output should look like as given below:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144,

```
#include <stdio.h>
int main() {
    int n = 12, i;
    int t1 = 1, t2 = 1, next;
    printf("%d, %d, ", t1, t2);
    for(i = 3; i <= n; i++) {
        next = t1 + t2;
        printf("%d, ", next);
        t1 = t2;
        t2 = next;
    }
    return 0;
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144,
-----
Process exited after 0.0896 seconds with return value 0
Press any key to continue . . .
```

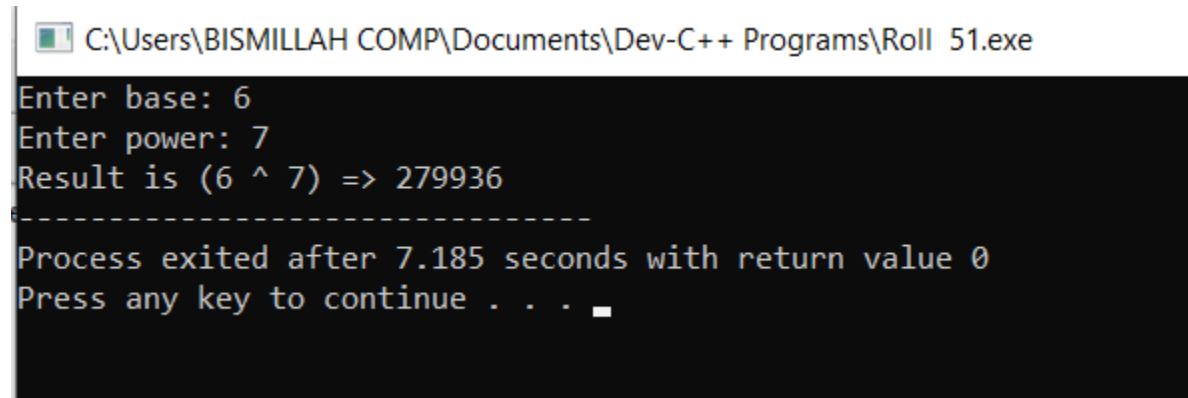
Task No. 7) Two numbers are entered through the keyboard. Write a program to find the value of one number raised to the power of another. You are not allowed to use pow() built in function.

Enter base: 3

Enter power: 4

Result is (3 ^ 4) => 81

```
#include <stdio.h>
int main() {
    int base, exp, i, result = 1;
    printf("Enter base: ");
    scanf("%d", &base);
    printf("Enter power: ");
    scanf("%d", &exp);
    for(i = 1; i <= exp; i++)
        result *= base;
    printf("Result is (%d ^ %d) => %d", base, exp, result);
    return 0;
}
```



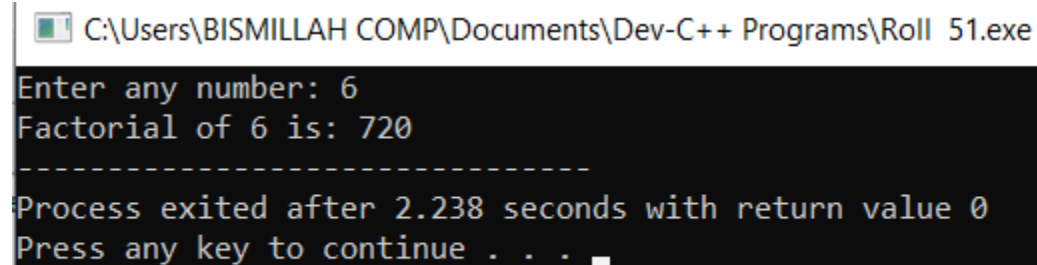
```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter base: 6
Enter power: 7
Result is (6 ^ 7) => 279936
-----
Process exited after 7.185 seconds with return value 0
Press any key to continue . . .
```

Task No. 8) Write a program to find the factorial value of any number entered through the keyboard.

Enter any number: 5

Factorial of 5 is: 120

```
#include <stdio.h>
int main() {
    int n, i;
    long fact = 1;
    printf("Enter any number: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
        fact *= i;
    printf("Factorial of %d is: %ld", n, fact);
    return 0;
}
```



```
C:\Users\BISMILLAH COMP\Documents\Dev-C++ Programs\Roll 51.exe
Enter any number: 6
Factorial of 6 is: 720
-----
Process exited after 2.238 seconds with return value 0
Press any key to continue . . .
```

THE END